

DATAFLOW — Project Brief

EXECUTIVE SUMMARY

DataFlow is a modern SaaS platform automating data pipeline development for mid-market companies. This portfolio project demonstrates full-stack capabilities: React frontend design, Node.js/Express backend architecture, responsive mobile-first CSS, and production-ready DevOps practices.

PROJECT GOALS

Primary: High-conversion landing page driving free trial signups. **Secondary:** Market differentiation vs. Airflow/dbt; responsive design across devices; SEO best practices; qualified email lead generation. **Success Metric:** 5% trial conversion, <2s load time, 40%+ mobile traffic, 15%+ trial-to-paid within 30 days.

KEY FEATURES DELIVERED

Frontend: React 18 components, CSS-in-JS styling (production: Tailwind), animated flow visualization, sticky navigation, form validation, WCAG 2.1 AA accessibility. **Design:** Dark navy (#0F172A) + teal (#06B6D4) color system, mobile-first responsive layout (375px–1920px), professional typography, conversion-optimized CTA hierarchy. **Sections:** Hero with animation, features grid, pricing tiers with toggle, testimonials, contact form, mobile views.

TECHNOLOGY STACK

Layer	Technology	Purpose
Frontend	React 18, CSS-in-JS, Vite	SPA with responsive design, smooth animations
Backend	Node.js, Express.js	REST API, routing, middleware, validation
Database	PostgreSQL	Transactional data: users, signups, contacts, analytics
Styling	Tailwind CSS (prod) / Inline CSS	(demo)Mobile-first responsive framework, design tokens
Deployment	Vercel + Railway/Heroku	Frontend CDN, backend containerization, auto scaling
CI/CD	GitHub Actions	Automated testing, staging deploy, production release
Third-Party	SendGrid, Stripe, Sentry, GA4	Email, payments, error tracking, analytics

4-WEEK DELIVERY TIMELINE

Week 1: Design system, repo setup, database schema. **Week 2:** Frontend components (Hero, Features, Pricing, Testimonials, Contact). **Week 3:** Express API, form integration, database queries. **Week 4:** QA/testing, Lighthouse optimization, production deployment.

DELIVERABLES

Code: React component library (12–15 components), Express API (8+ endpoints), PostgreSQL migrations, TypeScript typings, environment config. **Documentation:** README, API spec (OpenAPI), component storybook, deployment guide. **Performance:** Lighthouse 90+ score, <2s load, Core Web Vitals passing. **Extras:** Figma design file, video script + storyboard (52s), email templates, Google Analytics events setup, Sentry monitoring.

COMPETITIVE POSITIONING

vs. Airflow: No-code builder (not DAG code); managed hosting (not self-hosted). **vs. dbt:** End-to-end pipeline (not SQL-only); single platform (not warehouse-dependent). **vs. Fivetran:** Transparent per-record pricing (not per-connector); transformations on all plans (not premium-only).

PORTFOLIO VALUE

This project showcases **full-stack ownership**: component architecture, CSS discipline, API design, database modeling, DevOps setup. The **distinctive design system** (navy + teal, intentional typography, section rhythm) differentiates from template defaults. The **52-second demo video script** and **9-panel storyboard** demonstrate marketing alignment and communication skills—essential for full-stack developers driving customer acquisition and revenue impact.